PRODUCT_REQUIREMENTS_DOCUMENT.md

EduMentor AI

Product Requirements Document (PRD)

Version: 2.0

Status: MVP

Executive Summary

EduMentor AI is an AI-powered personalized learning platform designed to help students learn Python through conversational tutoring, adaptive quizzes, progress tracking, and personalized learning sessions.

Unlike generic AI chatbots, EduMentor AI focuses on structured learning by maintaining learning history, tracking mastery levels, adapting quiz difficulty, and providing personalized educational guidance based on the learner's progress.

The platform aims to provide every learner with access to a private, personalized AI tutor available 24/7.

Problem Statement

Many students struggle with programming education because:

- Traditional courses are static.
- Learning platforms cannot adapt to individual learners.
- Students often hesitate to ask questions repeatedly.
- Personalized tutoring is expensive.
- Progress is difficult to measure.
- Learners do not know what to study next.

As a result, students become overwhelmed, lose motivation, and stop learning.

Solution

EduMentor AI provides a personalized AI tutoring experience that:

- Explains Python concepts conversationally.
- Maintains persistent learning sessions.
- Remembers learning context.
- Generates adaptive quizzes.
- Tracks topic mastery.
- Identifies weak areas.
- Personalizes assessments based on performance.
- Enables learners to continue learning at any time.

The platform combines tutoring, assessment, and analytics into a unified educational experience.

Vision Statement

To make high-quality personalized programming education accessible to anyone through AI-powered tutoring and adaptive learning.

Target Audience

Primary Audience

Beginner Students

Users learning Python for the first time.

Examples:

- School Students
- College Students
- Self-Learners
- Career Switchers

Secondary Audience

Intermediate Learners

Users seeking:

- Concept clarification
- Practice questions
- Revision support
- Interview preparation

Goals

Business Goals

- Build a high-quality hackathon MVP.
- Demonstrate adaptive AI learning.
- Showcase personalization.
- Validate demand for AI tutoring.

User Goals

Users should be able to:

- Learn Python concepts.
- Ask unlimited questions.
- Receive personalized explanations.
- Take adaptive quizzes.
- Measure learning progress.
- Resume previous sessions.
- Improve weak topics.

Success Metrics

User Success

Users should be able to:

1. Register successfully.
2. Complete onboarding.

3. Start a learning session.
4. Learn through AI tutoring.
5. Generate a quiz.
6. Complete a quiz.
7. View progress updates.
8. Continue learning later.

Product Success

Target Metrics:

- Session Completion Rate > 80%
- Quiz Completion Rate > 70%
- Return User Rate > 40%
- Average User Satisfaction > 4/5

Scope

Included in MVP

Authentication

- Email Registration
- Email Login
- Session Management

User Onboarding

Collect:

- Learning Topic
- Skill Level
- Learning Goal
- Daily Study Time

AI Tutoring

- Conversational Learning
- Context-Aware Responses
- Personalized Explanations
- Session Memory

Learning Sessions

- Create Sessions
- Resume Sessions
- Rename Sessions
- Delete Sessions

Adaptive Quiz System

- Quiz Generation
- Dynamic Difficulty Selection
- Answer Evaluation
- Quiz History
- Performance Feedback

Progress Tracking

- Topic Mastery
- Weak Topic Detection
- Learning Analytics
- Performance Trends

Profile Management

- Update Profile
- Update Preferences

Settings

- Theme Preferences
- Learning History Preferences
- Account Management

Out of Scope (Phase 2)

The following features are intentionally excluded from the MVP:

- Voice Tutor
- JavaScript Learning Track

- File Uploads
- Code Execution Sandbox
- AI Roadmap Generator
- Multi-Language Learning
- Teacher Dashboard
- Parent Dashboard
- Certificates
- Study Groups
- Collaborative Learning
- Leaderboards

Core Features

Feature 1: AI Tutor

The AI Tutor provides:

- Personalized explanations
- Beginner-friendly responses
- Context retention
- Educational guidance

The tutor must adapt explanations based on:

- User skill level
- Learning goals
- Previous conversations

Feature 2: Persistent Learning Sessions

Users must be able to:

- Create learning sessions
- Continue previous sessions
- View learning history
- Organize conversations

All sessions must be stored securely.

Feature 3: Adaptive Learning Engine

The Adaptive Learning Engine personalizes quiz difficulty using mastery scores.

Mastery Levels

Mastery Score	Difficulty
0 - 40%	Beginner
41 - 70%	Intermediate
71 - 100%	Advanced

Quiz Adaptation Logic

Before generating a quiz:

1. Retrieve user progress.
2. Calculate mastery score.
3. Determine difficulty level.
4. Generate appropriate questions.

Example:

```
Mastery = 25%
Difficulty = Beginner
```

Example:

```
Mastery = 82%
Difficulty = Advanced
```

This ensures assessments remain personalized.

Feature 4: Progress Tracking

Track:

- Quiz Scores
- Topic Mastery
- Learning Time
- Session Activity

Users should understand:

- What they know

- What they struggle with
- What to study next

User Personas

Persona 1

Beginner Student

Name: Sarah

Age: 19

Goal:

Learn Python for university coursework.

Pain Points:

- Difficult tutorials
- Lack of support
- Fear of asking questions

Desired Outcome:

Understand Python confidently.

Persona 2

Career Switcher

Name: Alex

Age: 28

Goal:

Learn Python for software development jobs.

Pain Points:

- Limited time
- Need measurable progress

- Need structured learning

Desired Outcome:

Become job-ready.

Functional Requirements

Authentication

FR-001

Users must be able to register.

FR-002

Users must be able to login.

FR-003

Users must remain authenticated across sessions.

Onboarding

FR-004

Users must complete onboarding.

FR-005

User preferences must be stored.

Learning Sessions

FR-006

Users must create sessions.

FR-007

Users must view session history.

FR-008

Users must resume sessions.

FR-009

Users must rename sessions.

FR-010

Users must delete sessions.

AI Tutor

FR-011

Users must send messages.

FR-012

System must generate contextual responses.

FR-013

System must use conversation history.

FR-014

System must personalize explanations.

Quiz System

FR-015

Users must generate quizzes.

FR-016

Users must submit answers.

FR-017

System must score quizzes.

FR-018

System must store results.

Progress Tracking

FR-019

System must calculate mastery scores.

FR-020

System must identify weak topics.

FR-021

System must display learning analytics.

Adaptive Learning

FR-022

The system must dynamically adjust quiz difficulty based on mastery score.

FR-023

The system must use stored progress data when generating quizzes.

FR-024

The system must provide personalized assessments for each learner.

Security Requirements

The platform must:

- Protect user accounts.
- Protect learning sessions.
- Protect chat history.
- Protect quiz results.
- Protect progress analytics.

Users may only access their own data.

Non-Functional Requirements

Performance

Chat Response:

```
< 3 seconds
```

Quiz Generation:

```
< 5 seconds
```

Dashboard Load:

```
< 2 seconds
```

Reliability

The platform must:

- Persist sessions
- Persist quiz results
- Persist progress data

without data loss.

Security

Must use:

- Supabase Auth
- Row Level Security
- Protected API Routes
- Secure Environment Variables

Technical Requirements

Frontend

- Next.js 15
- TypeScript
- Tailwind CSS
- shadcn/ui
- Lucide React

Backend

- Next.js Route Handlers
- TypeScript

Database

- Supabase PostgreSQL
- Supabase pgvector

Authentication

- Supabase Auth

AI Layer

- Groq
- Llama 3.3 70B Versatile

Embeddings

- Transformers.js
- all-MiniLM-L6-v2 (384 dimensions)

Hosting

- Render

User Flow

Landing Page ↓ Register ↓ Onboarding ↓ Dashboard ↓ Learning Session ↓ AI Tutor ↓ Adaptive Quiz ↓ Progress Update ↓ Continue Learning

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Risks

Technical Risks

- AI Hallucinations
- Context Window Limits
- Poor Quiz Quality

Mitigation:

- Structured Prompting
- Session Context Management
- Adaptive Difficulty Logic

Product Risks

- Feature Scope Expansion
- Over-Engineering

Mitigation:

- Strict MVP Scope
- Focus on Tutoring + Quizzes

MVP Completion Criteria

The MVP is complete when users can:

1. Create an account.
2. Complete onboarding.
3. Start a learning session.
4. Chat with the AI tutor.
5. Generate an adaptive quiz.
6. Submit a quiz.
7. View progress updates.
8. Resume previous sessions.

Future Vision

EduMentor AI will evolve into a complete AI learning ecosystem featuring:

- Multiple Programming Languages
- Voice Tutoring
- AI Learning Roadmaps
- Interactive Coding Environments
- Teacher Dashboards
- Parent Dashboards
- Certifications
- Collaborative Learning

The MVP serves as the foundation for this long-term vision.